AI Health Screening Tool: Validation Results Report

Project: AI-Powered Health Screening for Rural Rajasthan **Investigators:** Duncan Webb (NovaSBE), Pascaline Dupas (Princeton), Patrick Agte (SSE), ICMR-NIRDHDS (Dr. Mona Duggal)
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Overview

This report documents the validation work conducted on our AI-powered health screening tool prior to and during deployment in a randomized controlled trial in rural Rajasthan, India. The tool uses a large language model (LLM) to conduct structured health screenings via community health workers, generating severity scores and health summaries for patients aged 40+ in underserved communities.

Our validation framework comprises three categories of evidence: (1) simulated clinical vignette testing against established medical benchmarks, (2) test-retest reliability analysis of severity score consistency, and (3) independent medical expert review of real screening transcripts from the field. We also describe validation phases that are in progress or planned.

This is a living document. Results will be updated as additional validation data become available.

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1. Clinical Vignette Simulations

1.1 Methodology

We developed an automated vignette simulation framework to benchmark our AI screening tool against clinical standards. The framework orchestrates conversations between an LLM-powered "Doctor Simulator" (using our screening prompt) and a scripted "Patient Simulator" that responds from standardized case definitions.

Case definitions. We adapted three clinical vignettes from Das et al. (2016, *American Economic Review*) — tuberculosis (TB), dysentery, and unstable angina — which were originally developed for standardized patient (SP) assessments of healthcare provider quality in rural India. The use of SP-based vignettes for measuring clinical competence was established by Das & Hammer (2005). We supplemented these with two additional vignettes (pre-eclampsia and asthma) to broaden coverage of conditions relevant to the target population. The five conditions span common and serious presentations. Each case includes a structured set of history and examination questions with predefined correct answers, an answer key specifying the correct diagnosis, and a list of correct, palliative, and unnecessary/harmful treatments.

Simulation procedure. The Doctor Simulator conducts the clinical interview following our screening prompt, dynamically asking follow-up questions based on patient responses. The Patient Sim-

ulator uses semantic similarity matching (cosine similarity, threshold 0.6) to map the doctor's free-text questions to the closest predefined question-answer pair, ensuring consistent patient information across simulations. Each simulation produces a full conversation transcript, an extracted diagnosis, and treatment recommendations.

Evaluation metrics. Simulated consultations were scored on three dimensions:

- **Checklist completion rate**: The proportion of clinically recommended questions asked, following the methodology of Das et al. (2016). These checklists define the essential history-taking and examination items for each condition.
- **Diagnostic accuracy**: Whether the LLM's extracted diagnosis matched the correct diagnosis from the answer key (classified as correct, partially correct, or incorrect by an independent LLM evaluator).
- **Treatment appropriateness**: Whether recommended treatments were classified as correct, palliative, or unnecessary/harmful based on the answer key.

Scale. We ran 998 simulated consultations across the five conditions, testing GPT-5.2 in both English and Hindi, and Gemini 2.5 Flash in English, with 20+ simulations per model-case combination.

1.2 Results

Overall performance. Across all models and conditions, the AI screening tool achieved:

- **87% checklist completion rate** (proportion of clinically recommended questions asked)
- **99% correct diagnosis rate**
- **99% appropriate treatment recommendations** (correct or palliative)

Comparison with real-world providers. Das et al. (2016) measured the quality of care delivered by actual healthcare providers in rural and urban India using standardized patient (SP) vignettes, finding that providers completed an average of 27% of recommended checklist items and achieved correct diagnoses in only 6% of cases. Das et al. (2022) document similarly low quality across primary care markets in rural Indian villages. Our AI tool substantially outperforms these benchmarks on information gathering and diagnostic accuracy.

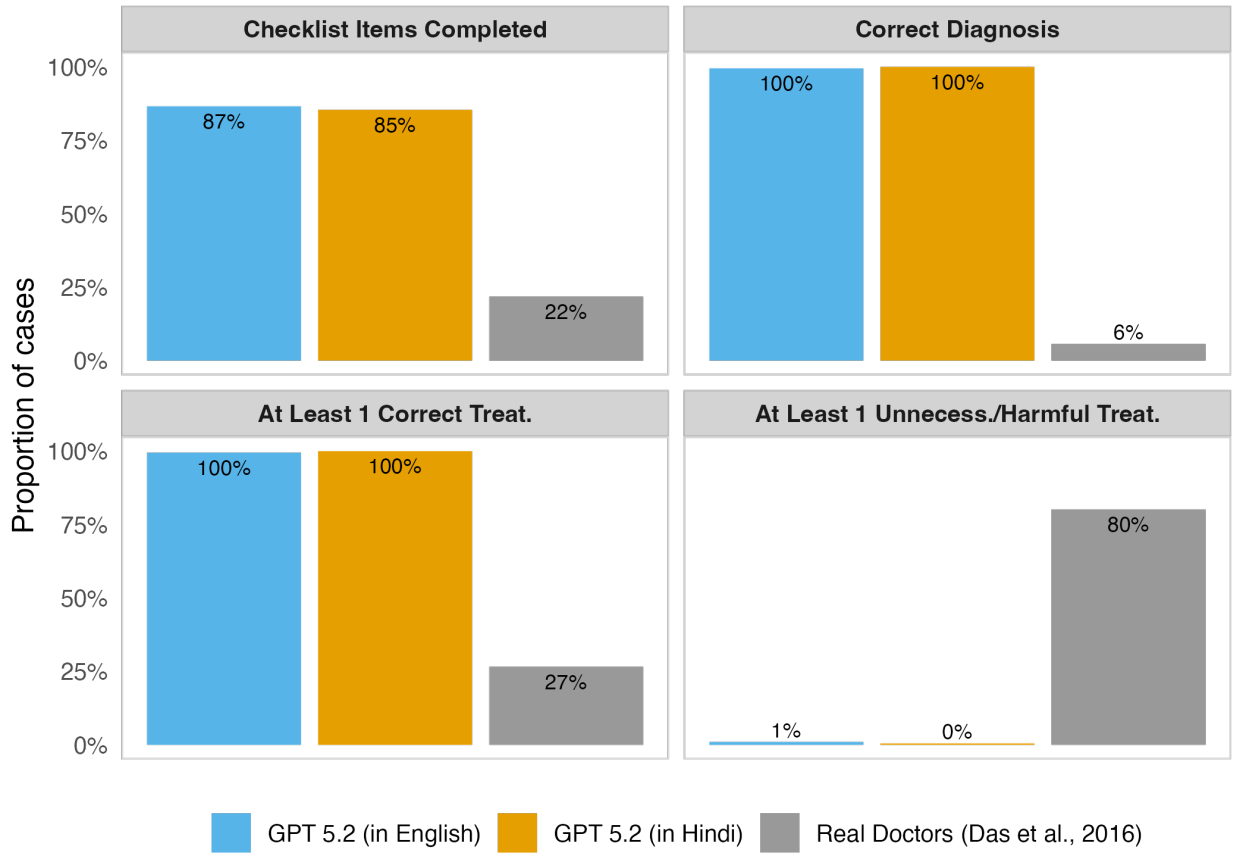

Figure 1. Comparison of AI model performance against real-world provider benchmarks from Das et al. (2016). The AI tool achieves substantially higher checklist completion and diagnostic accuracy across all models and languages tested.

Hindi language performance. GPT-5.2 achieved comparable performance in Hindi and English, confirming the tool's viability for deployment in Hindi-medium health screenings in rural India.

Question efficiency. We additionally tested varying question budgets (5, 8, 10, 12, and 15 questions) and adaptive modes where the model self-regulates its questioning depth. With as few as 8 questions, the model maintained high diagnostic accuracy, suggesting efficient information gathering even under time constraints relevant to field deployment.

1.3 Limitations

- Vignette patients respond from a fixed script; real patients provide less structured information.
- The semantic matching approach may overstate the AI's ability to interpret natural conversational responses.
- Cases are drawn from conditions well-represented in medical literature; performance on rarer or more ambiguous presentations has not been tested in simulation.

2. Test-Retest Reliability

2.1 Methodology

To assess the consistency of the AI tool's severity scoring, we conducted a test-retest reliability analysis. We selected 20 complete screening conversations from actual field deployment in rural Rajasthan (October–November 2025) and replayed each conversation through the OpenAI API 10 times, generating 200 independent severity score assessments.

The AI tool generates a **provider importance score** on a 0–10 scale for each patient ("How important is it that this patient see a healthcare provider?"). This is the primary score used in the intervention to determine whether patients receive health information and referral recommendations.

Each conversation was submitted to GPT-5.2 using the same system prompt used in the field. The model regenerated severity scores from the transcript. When the model asked additional questions beyond the transcript, a default response of "no further problems" was provided. Of 200 API calls, 190 were successful (10 errors due to exceeding conversation turn limits).

2.2 Results

Provider importance scores are consistent within +/-1 point of the original 84% of the time. The mean within-conversation standard deviation is 0.50 points on the 10-point scale, and the within-conversation SD after removing conversation fixed effects is 0.59, confirming that stochastic variation across repetitions is small relative to the between-conversation variation that drives clinical decisions.

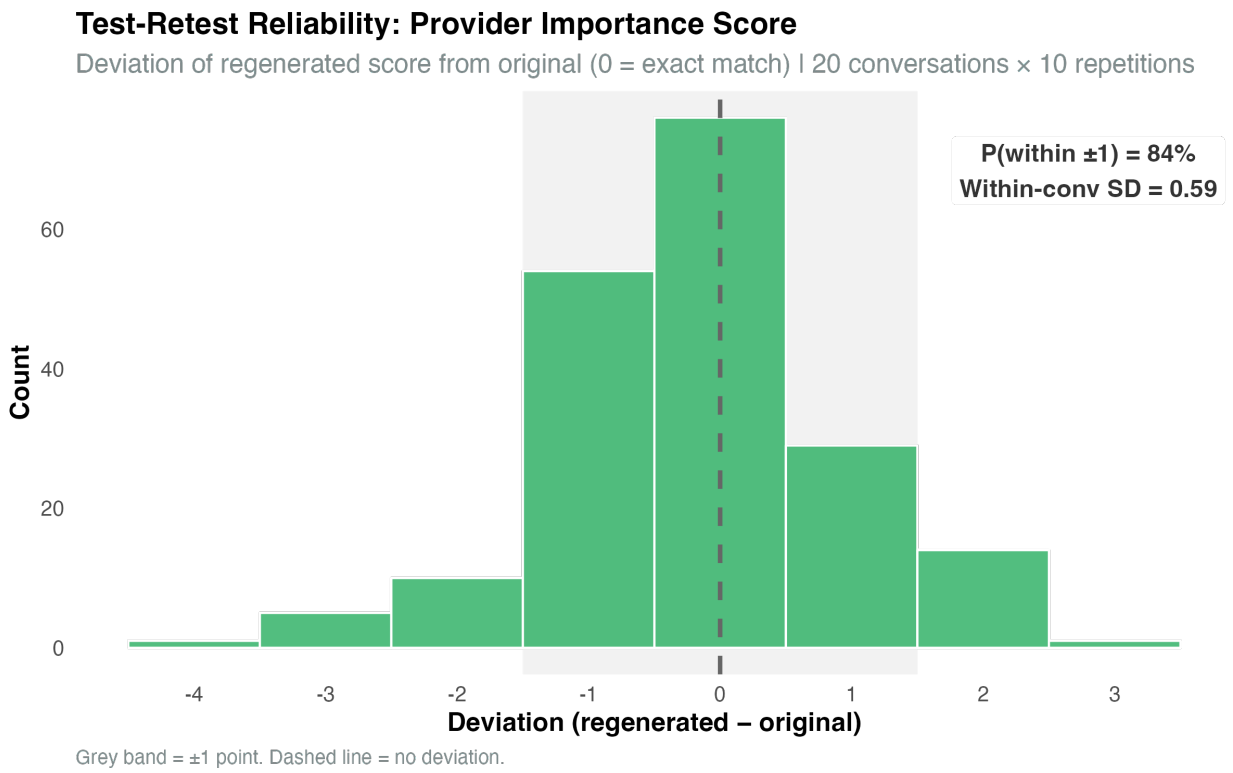

Figure 2. Distribution of regenerated provider importance score deviations from original scores. 84% of regenerated scores fall within +/-1 of the original. Grey band = +/-1 point.

2.3 Limitations

- The system prompt was assumed to match the Custom GPT configuration used in the field; minor differences may exist.
- The API model (GPT-5.2) differs slightly from the original field model (GPT-5.1 Instant).
- When extra conversation turns were needed, the default patient response may affect summaries.
- Sample is limited to 20 conversations from the pilot.

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3. Phase 1 Expert Review

3.1 Methodology

Four senior medical experts independently reviewed the AI screening prompt and 10 curated transcripts from field screening in rural Rajasthan (Udaipur district, October–November 2025). The expert panel was assembled by co-PI Dr. Mona Duggal (ICMR-NIRDHDS) and includes specialists in community medicine and maternal/child health, public health administration and digital health (including India's National NCD Portal), clinical practice in the study area, and Indian Public Health Standards (IPHS) and National Health Mission (NHM) implementation.

Each expert was asked to: (1) write a structured 5–10 page report evaluating the prompt's medical soundness, (2) provide annotated feedback on each transcript with severity ratings, and (3) participate in a 60-minute debrief call with the research team. This phase was unblinded — experts knew the outputs were AI-generated — and was designed to improve the prompt before the blinded Phase 2 validation.

3.2 Severity Score Agreement

Experts independently rated each transcript on a 0–10 severity scale ("How important is it that this patient see a healthcare provider?"). We compared these ratings against the AI tool's own provider importance scores.

The AI is largely unbiased relative to expert consensus, with a mean bias of +0.40 points (AI minus mean expert). The AI's severity scores fall within +/-1 point of the mean expert score for 80% of transcripts (100% when excluding the most conservative expert).

Comparison	Pearson r	n (pairs)
AI vs. expert	**0.85**	37
Expert vs. expert	**0.81**	28

The AI-expert correlation (r = 0.85) exceeds the inter-expert correlation (r = 0.81), indicating that the AI's severity assessments are at least as consistent with expert judgment as experts are with each other. Three of four experts show very high agreement with each other (pairwise r = 0.93–0.98) and with the AI (r = 0.94–0.99). One expert systematically rates acute cases (chest pain, breathlessness) as moderate (5/10) where the other experts and the AI rate them as urgent (8–10/10) — reflecting a clinical judgment difference about over-triage risk in resource-constrained settings.

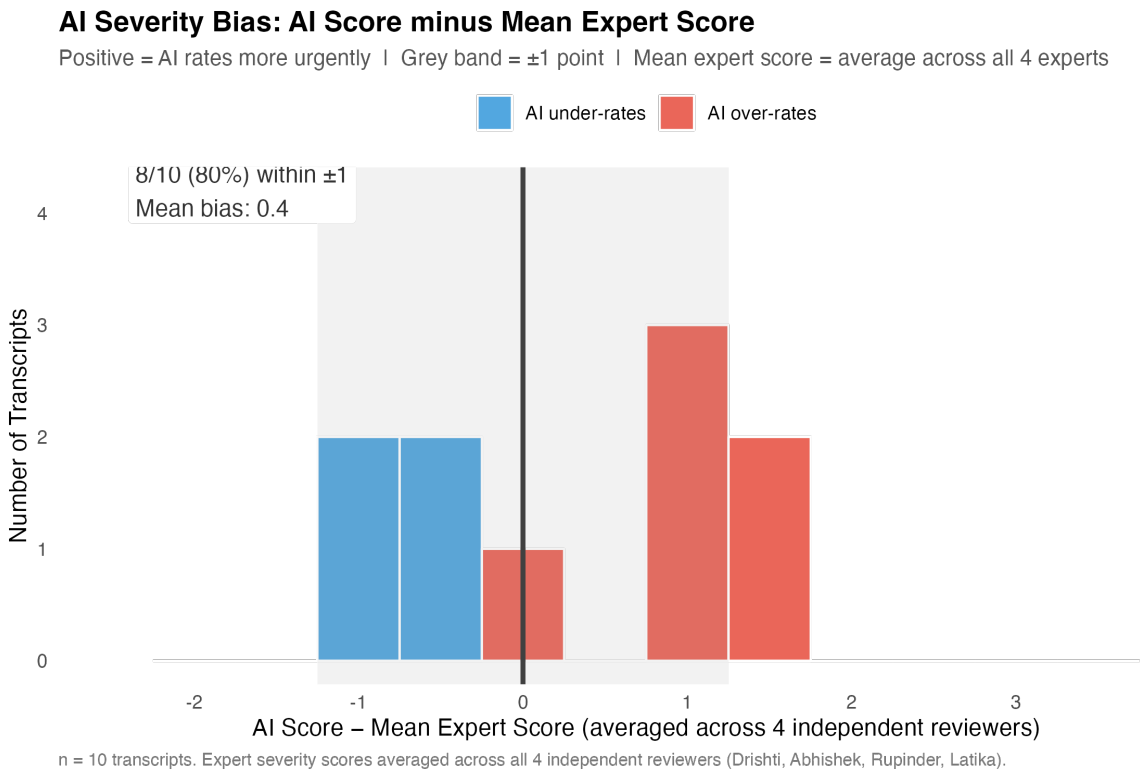

Figure 3. Distribution of AI severity bias (AI score minus mean expert score) across 10 transcripts. Mean expert score is the average severity rating across all 4 independent reviewers. 80% of transcripts fall within +/-1 point (grey band). The AI is largely unbiased (mean bias = +0.40).

3.3 Key Qualitative Findings

Areas of improvement identified by experts 1. The AI asks multiple questions simultaneously too often, losing information from elderly patients who respond only to the last question. 2. The AI should ask more about current medications, prior diagnoses, or previous care-seeking. 3. Referral recommendations must be realistic for rural healthcare infrastructure (aligned with Indian Public Health Standards facility tiers).

Actionable prompt improvements identified: emergency exit logic for acute presentations, one-question-at-a-time enforcement, symptom history probing (onset, duration, treatment), stepped-care referral model, and substance use screening for the Rajasthan context. Experts provided 431 inline annotations across all reviews.

3.4 Overall Expert Verdicts

- "Highly medically sound for a community-level intervention." Confidence: *high, conditional on safety guardrails*.
- "The AI tool carries huge potential in primary screening and referral." Confidence: *high, conditional on interviewer training*.

- "The potential is high… if medical history and medications are well recorded, the AI would be successful." Confidence: *high, conditional on deeper probing*.
- One expert found the tool promising but currently limited by over-diagnosis tendencies. Confidence: *limited*.

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4. Planned and In-Progress Validation

4.1 Phase 2: Blinded Transcript Validation (In Progress)

Status: Started March 23, 2026.

Design. Medical experts independently review screening transcripts without seeing the AI's outputs, providing their own severity assessments for comparison. Unlike Phase 1, this phase is blinded and uses a larger, representative sample of transcripts.

Sample: 300 screening transcripts for the primary model (GPT-5.2), selected through stratified random sampling across severity levels, patient demographics, and presenting complaint categories. An additional 100 transcripts will be processed through the secondary model (Google Gemini) for head-to-head comparison. Each transcript is reviewed by at least 2 independent medical experts.

Outcome measures:

- Agreement rate between AI and expert severity scores (within +/-1 point)
- False negative rate: proportion of high-severity cases (expert score >= 7) rated low by AI (score <= 4)
- False positive rate: proportion of low-severity cases (expert score <= 3) rated high by AI (score >= 7)
- Summary quality scores

Acceptance criteria:

- Agreement rate >= 80%
- False negative rate <= 5% (critical safety threshold)
- False positive rate <= 20%

Following any significant prompt modifications, a verification round of 50 additional transcripts will confirm improvements.

4.2 Phase 3: Live Field Observation (Planned)

Design. Medical experts will accompany field teams to observe 100 live screening conversations as they occur naturally. For each interaction:

1. The expert observes silently during the AI-mediated screening, noting conversation flow, questions asked, and patient responses.

2. After the screening, the expert conducts a brief independent clinical assessment of the patient (5–10 minutes).
3. Expert findings are compared with AI-generated outputs to evaluate real-world performance.

Assessment criteria: Completeness of symptom elicitation, accuracy of patient response interpretation, severity score calibration against clinical judgment, recommendation actionability, and whether any concerning symptoms were missed.

4.3 Red Teaming (Planned)

Prior to main trial launch, research assistants and medical consultants will attempt to elicit failures from the AI tool in targeted scenarios: ambiguous or conflicting symptom presentations; acute emergency presentations (chest pain, stroke symptoms) where appropriate escalation is critical; boundary cases where the tool should decline to provide diagnoses or treatment recommendations; culturally sensitive topics (reproductive health, mental health); contradictory or incomplete patient responses; and Hindi comprehension across regional dialect variations. The exercise will produce a documented log of identified failure modes and corresponding prompt fixes, with re-testing to confirm that critical issues have been resolved.

4.4 Ongoing Monitoring During Main RCT

Throughout the main study, continuous monitoring will be maintained:

- **Weekly transcript review**: 20–30 randomly selected transcripts reviewed by the ICMR-NIRDHDS medical expert panel.
- **Escalation protocol**: Flagged transcripts trigger immediate senior expert review, participant follow-up if clinically indicated, and root cause analysis.
- **Algorithmic fairness audits**: Regular checks on AI performance across gender, age, and caste to detect systematic bias.

References

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This is a living document. Updated validation results will be posted as they become available.